Experiment 5: Transistor Small-Signal Amplifiers

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5.A Objective

The goal of this experiment is to study small-signal transistor amplifiers built around the 2N2222 BJT. A common-emitter (CE) amplifier with and without an emitter bypass capacitor and a common-collector (emitter follower) amplifier are designed, simulated, and built. For each circuit the Q-point, maximum unclipped output, small-signal voltage gains $A_{\text{vi}} = \frac{v_o}{v_i}$ and $A_{\text{vs}} = \frac{v_o}{v_s}$, and the input and output resistances are calculated, then compared against simulation and laboratory measurements.

5.B Pre-Lab

Given values: $V_{\text{CC}} = 12V$, $R_C = 6.2k\Omega$, $R_E = 1.8k\Omega$, $R_L = 2.2k\Omega$, $R_S = 1.8k\Omega$, $C_E = 100\mu F$, $C_B = C_L = 10\mu F$. For the 2N2222 the lab manual gives $\beta \approx 150$ and $V_A \approx 100V$. Throughout, $V_{\text{BE}} = 0.7V$, $V_T = kT/q \approx 26mV$ and $V_{\text{CE(sat)}} \approx 0.2V$ are assumed.

5.B.1 Common-Emitter Amplifier

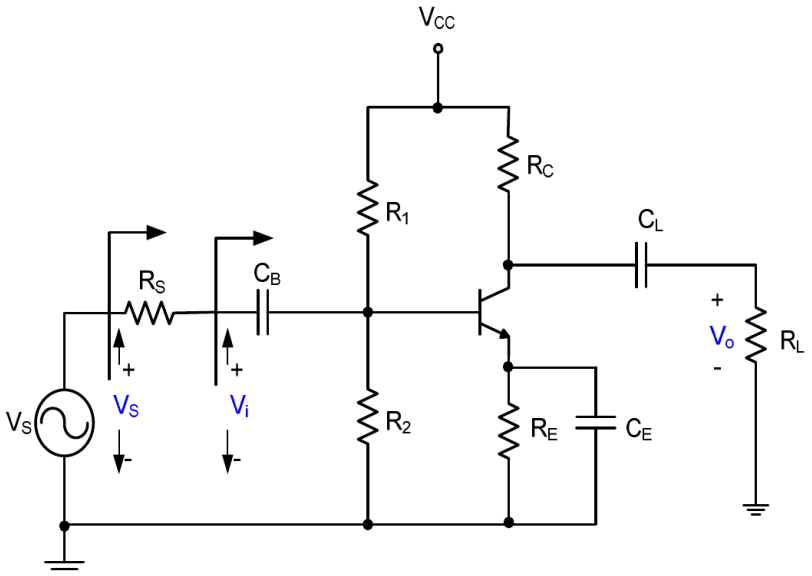


Figure 1: Common-emitter amplifier

5.B.1.a Calculate R_1 and R_2

Targeting $I_{\text{CQ}} \approx 1mA$

$$R_{\text{TH}} = .1 \times (1+\beta) \times R_E \\ = .1 \times 151 \times 1.8k\Omega \approx 27.2k\Omega \quad \text{Eq 1}$$

$$I_B = \frac{1mA}{150} = 6.67\mu A \\ I_E = I_B \times (1+\beta) = 1.007mA \quad \text{Eq 2}$$

$$V_{\text{TH}} = V_{\text{BE}} + I_E \times R_E + I_B \times R_{\text{TH}} \\ = .7 + 1.81 + .18 = 2.69V \quad \text{Eq 3}$$

$$V_{\text{TH}} = V_{\text{CC}} \times \frac{R_2}{R_1 + R_2} \\ R_1 = 121.249k\Omega \approx 120k\Omega \\ R_2 = 35.033k\Omega \approx 36k\Omega \quad \text{Eq 4}$$

5.B.1.b Compute Q-point and max unclipped output voltage

$$R_{\text{TH}} = R_1 \parallel R_2 = 120k\Omega \parallel 36k\Omega = 27.69k\Omega$$
$$V_{\text{TH}} = 12 \times \frac{36k}{120k + 36k} = 2.769V$$
Eq 5

$$I_{\text{BQ}} = \frac{V_{\text{TH}} - V_{\text{BE}}}{R_{\text{TH}} + (1+\beta)R_E} = \frac{2.769 - .7}{27.69k + 151 \times 1.8k} = 6.91\mu A$$
$$I_{\text{CQ}} = \beta I_{\text{BQ}} = 1.036mA, \quad I_{\text{EQ}} = (1+\beta)I_{\text{BQ}} = 1.043mA$$
Eq 6

$$V_{\text{CEQ}} = V_{\text{CC}} - I_{\text{CQ}}R_C - I_{\text{EQ}}R_E$$
$$= 12 - 1.036m \times 6.2k - 1.043m \times 1.8k = 3.70V$$
Eq 7

The Q-point is therefore $\left(I_{\text{CQ}}, V_{\text{CEQ}}\right) = (1.04mA, 3.70V)$.

$$R_{\text{ac}} = R_C \parallel R_L = 6.2k \parallel 2.2k = 1.624k\Omega$$
Eq 8

to find the maximum unclipped output

$$\text{Saturation limit:} \quad V_{\text{CEQ}} - V_{\text{CE(sat)}} = 3.70 - 0.2 = 3.50V$$
$$\text{Cutoff limit:} \quad I_{\text{CQ}}R_{\text{ac}} = 1.036m \times 1.624k = 1.68V$$
Eq 9

The cutoff limit is smaller, so the maximum unclipped output is $\approx 3.37V_{\text{pp}}$.

5.B.1.c Compute the small-signal voltage gain $A_{\text{vi}} = \frac{v_o}{v_i}$ and $A_{\text{vs}} = \frac{v_o}{v_s}$

$$g_m = \frac{I_{\text{CQ}}}{V_T} = \frac{1.036m}{26m} = 39.86mA/V$$
$$r_\pi = \frac{\beta}{g_m} = \frac{150}{39.86m} = 3.763k\Omega$$
$$r_o = \frac{V_A}{I_{\text{CQ}}} = \frac{100}{1.036m} = 96.5k\Omega$$
Eq 10

$$A_{\text{vi}} = -g_m(r_o \parallel R_C \parallel R_L) = -39.86m \times 1.597k = -63.7$$
$$A_{\text{vs}} = A_{\text{vi}} \times \frac{R_i}{R_i + R_S} = -63.7 \times \frac{3.313k}{3.313k + 1.8k} = -41.2$$
Eq 11

5.B.1.d Compute the input and output resistances

$$R_i = R_1 \parallel R_2 \parallel r_\pi = 27.69k \parallel 3.763k = 3.31k\Omega$$
$$R_o = R_C \parallel r_o = 6.2k \parallel 96.5k = 5.83k\Omega$$
Eq 12

5.B.2 Common-Emitter Amplifier without C_E

5.B.2.a Compute Q-point and max unclipped output voltage

C_E is an open circuit at DC, so removing it does not change the bias: the Q-point is still $(I_{\text{CQ}}, V_{\text{CEQ}}) = (1.04mA, 3.70V)$.

$$R_{\text{ac}} = R_C \parallel R_L + R_E = 1.624k + 1.8k = 3.424k\Omega \quad \text{Eq 13}$$

to find max swing

$$\begin{aligned} \text{Saturation:} \quad & i_{\text{c,max}} = \frac{V_{\text{CEQ}} - V_{\text{CE(sat)}}}{R_{\text{ac}}} = \frac{3.70 - 0.2}{3.424k} = 1.021mA \rightarrow v_o = 1.021m \times 1.624k = 1.66V \\ \text{Cutoff:} \quad & i_{\text{c,max}} = I_{\text{CQ}} = 1.036mA \rightarrow v_o = 1.036m \times 1.624k = 1.68V \end{aligned} \quad \text{Eq 14}$$

Saturation is reached first, so the maximum unclipped output is $\approx 3.32V_{\text{pp}}$.

5.B.2.b Compute the small-signal voltage gain Avi = vo/vi and Avs = vo/vs

The resistance looking into the base now includes the reflected emitter resistor:

$$R_{\text{ib}} = r_\pi + (1+\beta)R_E = 3.763k + 151 \times 1.8k = 275.6k\Omega \quad \text{Eq 15}$$

$$\begin{aligned} A_{\text{vi}} &= \frac{-\beta(R_C \parallel R_L)}{r_\pi + (1+\beta)R_E} = \frac{-150 \times 1.624k}{275.6k} = -0.884 \\ A_{\text{vs}} &= A_{\text{vi}} \times \frac{R_i}{R_i + R_S} = -0.884 \times \frac{25.16k}{25.16k + 1.8k} = -0.825 \end{aligned} \quad \text{Eq 16}$$

5.B.2.c Compute the input and output resistances

$$\begin{aligned} R_i &= R_1 \parallel R_2 \parallel [r_\pi + (1+\beta)R_E] = 27.69k \parallel 275.6k = 25.2k\Omega \\ R_o &\approx R_C = 6.2k\Omega \end{aligned} \quad \text{Eq 17}$$

5.B.3 Common-Collector (Emitter Follower) Amplifier

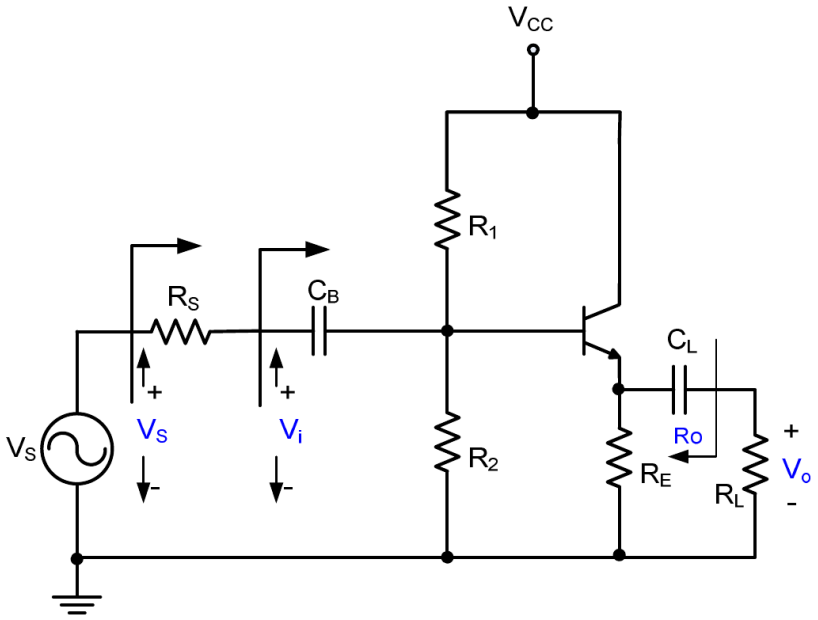


Figure 2: Emitter follower

5.B.3.a Calculate R_1 and R_2

$$R_{\text{TH}} = .1 \times (1+\beta) \times R_E = .1 \times 151 \times 1.8k\Omega \approx 27.2k\Omega$$
$$V_{\text{TH}} = V_{\text{BE}} + I_E \times R_E + I_B \times R_{\text{TH}} = .7 + 1.81 + .18 = 2.69V$$

Eq 18

$$R_1 = R_{\text{TH}} \times \frac{V_{\text{CC}}}{V_{\text{TH}}} = 121.2k\Omega \approx 120k\Omega$$
$$R_2 = R_{\text{TH}} \times \frac{V_{\text{CC}}}{V_{\text{CC}} - V_{\text{TH}}} = 35.0k\Omega \approx 36k\Omega$$

Eq 19

The same $R_1 = 120k\Omega$ and $R_2 = 36k\Omega$ were therefore used for both circuits.

5.B.3.b Compute Q-point and max unclipped output voltage

$$I_{\text{BQ}} = 6.91\mu A, \quad I_{\text{CQ}} = 1.036mA, \quad I_{\text{EQ}} = 1.043mA$$
$$V_E = I_{\text{EQ}} R_E = 1.043m \times 1.8k = 1.88V$$
$$V_{\text{CEQ}} = V_{\text{CC}} - I_{\text{EQ}} R_E = 12 - 1.88 = 10.12V$$

Eq 20

The Q-point is $(I_{\text{CQ}}, V_{\text{CEQ}}) = (1.04mA, 10.1V)$.

$$\text{Saturation limit:} \quad V_{\text{CEQ}} - V_{\text{CE(sat)}} = 10.12 - 0.2 = 9.92V$$
$$\text{Cutoff limit:} \quad I_{\text{EQ}} R_{\text{ac}} = 1.043m \times 990 = 1.03V$$

Eq 21

The maximum unclipped output is $\approx 2.07V_{\text{pp}}$, limited by cutoff.

5.B.3.c Compute the small-signal voltage gain Avi = vo/vi and Avs = vo/vs

The effective emitter load is $R'_E = R_E \parallel R_L \parallel r_o = 1.8k \parallel 2.2k \parallel 96.5k = 980\Omega$.

$$A_{\text{vi}} = \frac{(1+\beta)R'_E}{r_\pi + (1+\beta)R'_E} = \frac{151 \times 980}{3.763k + 151 \times 980} = \frac{147.97k}{151.73k} = 0.975$$
$$A_{\text{vs}} = A_{\text{vi}} \times \frac{R_i}{R_i + R_S} = 0.975 \times \frac{23.42k}{23.42k + 1.8k} = 0.906$$

Eq 22

5.B.3.d Compute the input and output resistances

$$R_i = R_1 \parallel R_2 \parallel [r_\pi + (1+\beta)R'_E] = 27.69k \parallel 151.7k = 23.4k\Omega$$
$$R_o = \frac{r_\pi + R_1 \parallel R_2 \parallel R_S}{1+\beta} \parallel R_E \parallel r_o$$
$$= \frac{3.763k + 1.690k}{151} \parallel 1.8k \parallel 96.5k = 36.1 \parallel 1.8k \parallel 96.5k = 35.4\Omega$$

Eq 23

5.C Simulation

5.D Experimental Procedure

The circuits of Figure 1 and Figure 2 were built on a breadboard with $R_1 = 120k\Omega$, $R_2 = 36k\Omega$, $C_B = C_L = 10\mu F$ and $C_E = 100\mu F$, powered from a 12 V supply.

5.D.1 Common-Emitter Amplifier

Q-point: $I_{\text{CQ}} = 1.11mA$, $V_{\text{CEQ}} = 2.37V$. **Gains:** $v_s = 7.0mV_{\text{rms}}$, $v_i = 5.0mV_{\text{rms}}$, $v_o = 308mV_{\text{rms}}$

$$A_{\text{vs}} = -\frac{308m}{7.0m} = -44.0, \quad A_{\text{vi}} = -\frac{308m}{5.0m} = -61.6$$

Eq 24

The negative sign comes from the oscilloscope captures (Figure 3, Figure 4), where the output is 180° out of phase with the input.

Input resistance: $V_{R_S} = 2.3mV_{\text{rms}}$

$$i_s = \frac{2.3mV}{1.8k\Omega} = 1.28\mu A, \quad R_i = \frac{v_i}{i_s} = \frac{5.0mV}{1.28\mu A} = 3.91k\Omega$$

Eq 25

Output resistance: $v_L = 308mV_{\text{rms}}$ (with R_L), $v_o = 960mV_{\text{rms}}$ (without R_L)

$$R_o = R_L \times \frac{v_o - v_L}{v_L} = 2.2k \times \frac{960 - 308}{308} = 4.66k\Omega$$

Eq 26

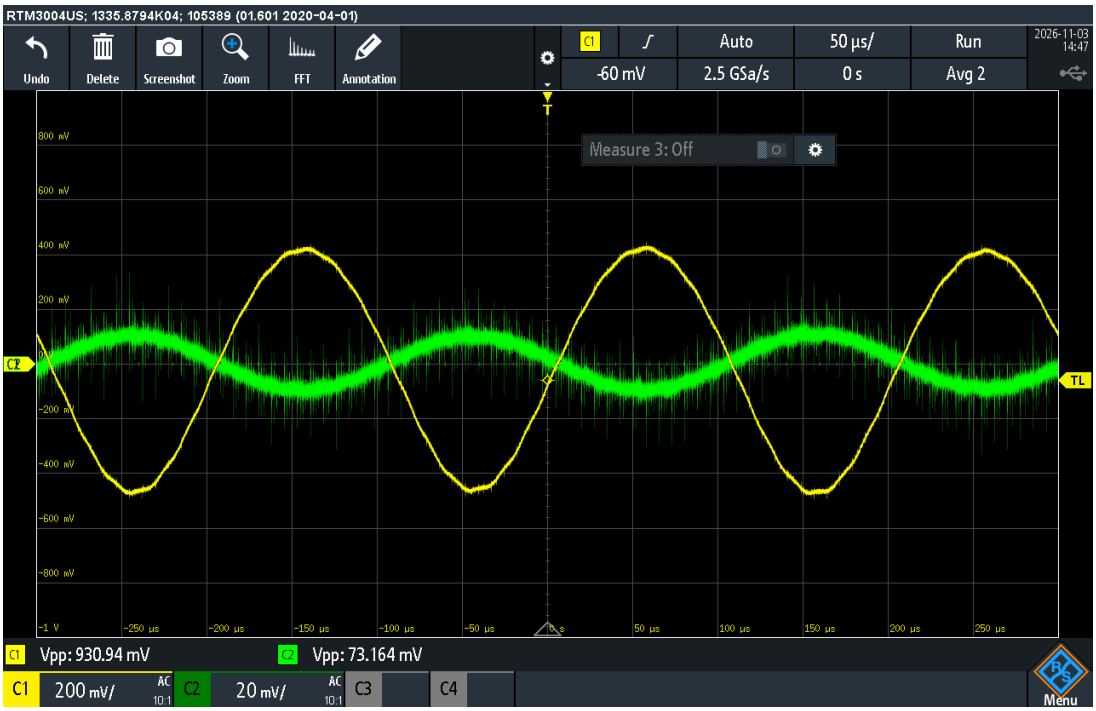


Figure 3: CE amplifier, v_s (CH2, green, 20 mV/div) and v_o (CH1, yellow, 200 mV/div) at 5 kHz

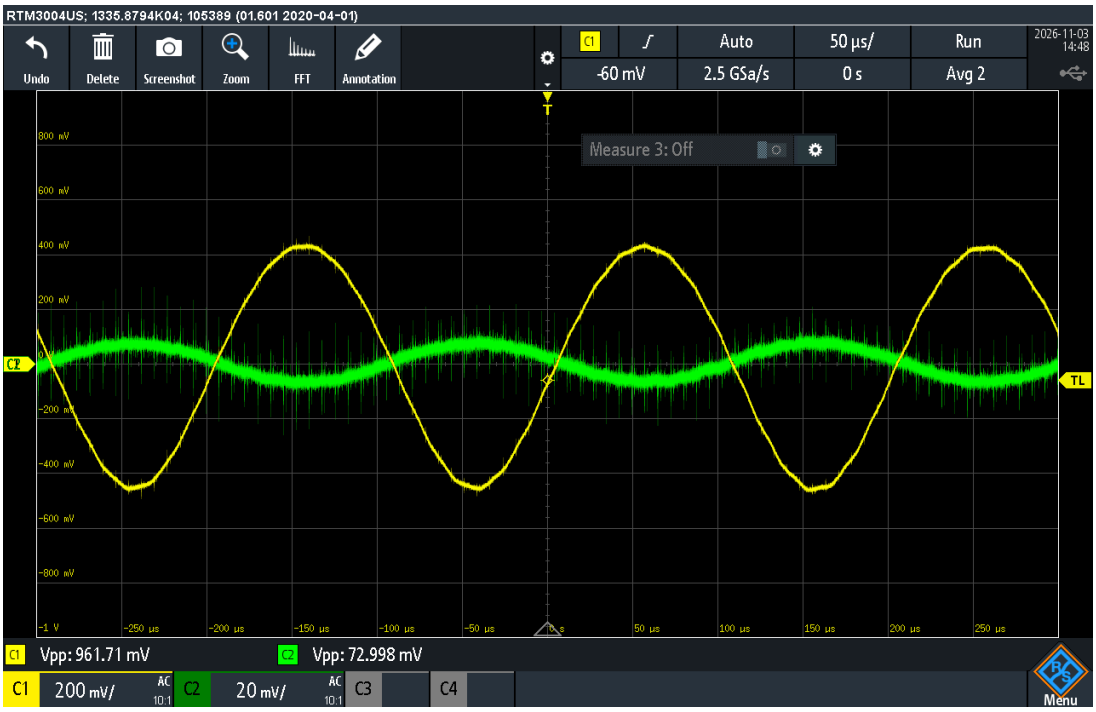


Figure 4: CE amplifier, v_i (CH2, green, 20 mV/div) and v_o (CH1, yellow, 200 mV/div) at 5 kHz

5.D.2 Common-Collector (Emitter Follower) Amplifier

Gains: $v_s = 7.0mV_{\text{rms}}$, $v_i = 6.6mV_{\text{rms}}$, $v_o = 6.4mV_{\text{rms}}$

$$A_{\text{vs}} = \frac{6.4m}{7.0m} = 0.914, \quad A_{\text{vi}} = \frac{6.4m}{6.6m} = 0.970 \qquad \text{Eq 27}$$

Input resistance: $V_{R_S} = 1.3mV_{\text{rms}}$

$$i_s = \frac{1.3mV}{1.8k\Omega} = 0.722\mu A, \quad R_i = \frac{v_i}{i_s} = \frac{6.6mV}{0.722\mu A} = 9.14k\Omega \tag{Eq 28}$$

Output resistance: $v_L = 6.4mV_{\text{rms}}$ (with R_L), $v_o = 6.5mV_{\text{rms}}$ (without R_L)

$$R_o = R_L \times \frac{v_o - v_L}{v_L} = 2.2k \times \frac{6.5 - 6.4}{6.4} = 34.4\Omega \tag{Eq 29}$$